

COP2800C Module 5 Graded Programming Assignment

In this assignment, we will build on the module content by modifying the Module 4 implementation of Palmer Penguins analysis to store and process data using arrays. The `readFile` method in the `CSVReader` class has been updated to read the entire data file and store the contents of a single column in a `String` array returned by the `readFile` method.

Instructions

1. Fork the public starter repository.

- Open the `PalmerPenguinsM5.java` file in your copy of the repo and complete the author and submission date lines in the ID header

2. Find the `CSVReader` class and note the changes to the `readFile` method

- a column number is now provided as a parameter to the method signature, and the method now returns a `String` array containing the values of the specified column instead of reading the data row by row. You do not need to make any other changes to this class or method.

```
public static String[] readFile(String FILE_NAME, int column) {
```

3. Complete the `TODO` items. Do not remove the `TODO` marker comments.

- **TODO 1:**
Declare a `String` array named `speciesData` which stores the results of calling `CSVReader.readFile` to read column 1 from the data file. Replace the ellipsis (...) with the array declaration.
- **TODO 2:**
Declare an `int` array named `speciesCount` with a size declarator of `NUM_SPECIES`.
- **TODO 3:**
Encode an `if` statement to determine if the length of the `speciesData` array is 0
- **TODO 4:**
Encode an enhanced `for` loop (also known as a `foreach` loop which iterates through the `speciesData` array using a loop variable named `species` of type `String`. Replace the ellipsis (...) with the `for` loop header.
- **TODO 5:**
Print the value of each `speciesCount` array element using the index constants. Use three separate `System.out.println` statements.

Example code:

```
System.out.println(SP_CHINSTRAP + " count = " +  
    speciesCount[SP_CHINSTRAP_INDEX]);
```

4. Test Your Program

- Run your program and verify that it correctly outputs the species counts.

Chinstrap count = 68

Gentoo count = 123

Adelie count = 151

5. Screenshot Your Output

- Take a screenshot of your program's output in the "Run I/O" pane of the jGrasp IDE.
- **DO NOT** the source code in the screenshot,

6. Submission

- a. Submit your completed PalmerPenguinsM5.java file to your GitHub repository.
 - **Do not** include .class files.
 - Ensure your code follows Java coding conventions (indentation, comments, line length).
- b. Upload your screenshot to the repository.
- c. Copy and paste your repository URL to the Canvas assignment as a comment.